GradeSense — Graded Answer Paper

Question 1 — Electric Circuit Principles

Student: Ananya Rao · Roll: 24B · Subject: Science

3.5 / 5

Evaluated

100% confidence

GradeSense - Answer Sheet

Name: Ananya Rao   Roll No: 24B

Q1. Science (5 marks)

An electric circuit is a close path in which electric current can flow. The battery gives the potentiall diffrence which push the current around the circuit. Switch is used for open and close the circuit, if switch is close then current will flow and bulb will glow, if switch is open then no current flow through the circuit.

In the circuit the battery, switch, resistor, bulb and ammeter is connected in series because current pass through all of them one by one. Ammeter is connected in series because it measure the current flowing in circuit. Voltmeter is also connected in the circuit to measure the potential diffrence, as shown in diagram below - 1

When current pass through the circuit some of the current get used up by the bulb and resistor, that is why the bulb glow and become hot.

As per Ohms law, V = IR. So if we increase the resistance then the current flowing in the circuit will also increase, because more resistance need more current to push through it. If the resistance is less then the bulb will glow dim. 2

Rubric Breakdown

Score: 3.5 / 5

- **Main circuit components (battery, switch, bulb, resistor) connected correctly in series** **1/1**
  Correctly identifies that the main components (battery, switch, resistor, bulb) are connected in series.

- **[1] Ammeter connected in series and voltmeter connected in parallel across the bulb (or component being** **0.5/1**
  Correctly states the ammeter is connected in series. However, it fails to specify that the voltmeter must be connected in parallel across the component being measured.

- **Explains closed path of current flow and functional roles of components** **1/1**
  Clearly explains the concept of a closed path for current flow and the functional roles of the battery and switch.

- **[2] Explains Ohm's law relationship (increased resistance decreases current flowing when voltage is const** **0/1**
  While Ohm's Law (V=IR) is stated, the explanation of its relationship between resistance and current is incorrect. Increasing resistance *decreases* current for a constant voltage, not increases it. The statement 'If the resistance is less then the bulb will glow dim' also contradicts the correct relationship.

- **Clear, logically structured explanation with a properly labelled diagram and conventional current direction** **1/1**
  AI visual assessment (from the uploaded page image): The diagram is properly labelled with all components and includes an arrow indicating the conventional direction of current flow, satisfying the criterion.